```
#include <stdio.h>
#include <string.h>

#define MAX_EMPLOYEES 100

struct Employee {
   int employeeID;
   char name[50];
   char department[50];
   char position[50];
   float basicSalary;
   float allowance;
   float deduction;
};

struct Employee employees[MAX_EMPLOYEES];
int totalEmployees = 0;

// Function declarations
void addEmployee();
void displayEmployees();
void searchEmployee();
void calculateSalary();
void employeeDetails();

int main() {
   int choice;
```

```
do {
    printf("\n====================================\n");
    printf(" MUNICIPAL EMPLOYEE MANAGEMENT SYSTEM\n");
    printf("====================================\n");
    printf("1. Add Employee\n");
    printf("2. Display All Employees\n");
    printf("3. Search for Employee\n");
    printf("4. Calculate Employee Salary\n");
    printf("5. Display Employee Details\n");
    printf("6. Exit\n");

    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            addEmployee();
            break;

        case 2:
            displayEmployees();
            break;

        case 3:
            searchEmployee();
            break;

        case 4:
            calculateSalary();
```

```
            break;

        case 5:
            employeeDetails();
            break;

        case 6:
            printf("\nExiting Employee Management System.\n");
            break;

        default:
            printf("\nInvalid choice. Please try again.\n");
      }

   } while (choice != 6);

   return 0;
}

// Function to add a new employee
void addEmployee() {

   if (totalEmployees >= MAX_EMPLOYEES) {
      printf("\nEmployee limit reached!\n");
      return;
   }

   printf("\n--- Add New Employee ---\n");
```

```
printf("Enter Employee ID: ");
scanf("%d", &employees[totalEmployees].employeeID);

// Check for duplicate employee ID
for (int i = 0; i < totalEmployees; i++) {
   if (employees[i].employeeID ==
      employees[totalEmployees].employeeID) {
      printf("This employee ID already exists.\n");
      return;
   }
}

printf("Enter Employee Name: ");
scanf(" %49[^\n]", employees[totalEmployees].name);

printf("Enter Department: ");
scanf(" %49[^\n]", employees[totalEmployees].department);

printf("Enter Position: ");
scanf(" %49[^\n]", employees[totalEmployees].position);

printf("Enter Basic Salary: ");
scanf("%f", &employees[totalEmployees].basicSalary);

printf("Enter Allowance: ");
scanf("%f", &employees[totalEmployees].allowance);

printf("Enter Deduction: ");
scanf("%f", &employees[totalEmployees].deduction);
```

```
    if (employees[totalEmployees].basicSalary < 0 ||
        employees[totalEmployees].allowance < 0 ||
        employees[totalEmployees].deduction < 0) {
        printf("Salary values cannot be negative.\n");
        return;
    }

    totalEmployees++;

    printf("\nEmployee successfully registered!\n");
}

// Function to display all employees
void displayEmployees() {

    if (totalEmployees == 0) {
        printf("\nNo employees have been registered.\n");
        return;
    }

    printf("\n--- Registered Employees ---\n");

    for (int i = 0; i < totalEmployees; i++) {

        printf("\nEmployee %d\n", i + 1);
        printf("Employee ID: %d\n", employees[i].employeeID);
        printf("Name: %s\n", employees[i].name);
        printf("Department: %s\n", employees[i].department);
```

```
        printf("Position: %s\n", employees[i].position);
        printf("Basic Salary: %.2f\n", employees[i].basicSalary);
    }
}

// Function to search for an employee
void searchEmployee() {

    int searchID;
    int found = 0;

    printf("\nEnter Employee ID to search: ");
    scanf("%d", &searchID);

    for (int i = 0; i < totalEmployees; i++) {

        if (employees[i].employeeID == searchID) {

            printf("\nEmployee Found!\n");
            printf("Employee ID: %d\n", employees[i].employeeID);
            printf("Name: %s\n", employees[i].name);
            printf("Department: %s\n", employees[i].department);
            printf("Position: %s\n", employees[i].position);
            printf("Basic Salary: %.2f\n", employees[i].basicSalary);

            found = 1;
            break;
        }
    }
```

```
    if (found == 0) {
        printf("\nEmployee not found.\n");
    }
}

// Function to calculate employee salary
void calculateSalary() {

    int searchID;
    int found = 0;
    float netSalary;

    printf("\nEnter Employee ID: ");
    scanf("%d", &searchID);

    for (int i = 0; i < totalEmployees; i++) {

        if (employees[i].employeeID == searchID) {

            netSalary = employees[i].basicSalary
                    + employees[i].allowance
                    - employees[i].deduction;

            printf("\n--- Salary Information ---\n");
            printf("Employee Name: %s\n", employees[i].name);
            printf("Basic Salary: %.2f\n", employees[i].basicSalary);
            printf("Allowance: %.2f\n", employees[i].allowance);
            printf("Deduction: %.2f\n", employees[i].deduction);
```

```
            printf("Net Salary: %.2f\n", netSalary);

            found = 1;
            break;
        }
    }

    if (found == 0) {
        printf("\nEmployee not found.\n");
    }
}

// Function to display complete employee details
void employeeDetails() {

    int searchID;
    int found = 0;

    printf("\nEnter Employee ID: ");
    scanf("%d", &searchID);

    for (int i = 0; i < totalEmployees; i++) {

        if (employees[i].employeeID == searchID) {

            printf("\n===== EMPLOYEE DETAILS =====\n");
            printf("Employee ID: %d\n", employees[i].employeeID);
            printf("Employee Name: %s\n", employees[i].name);
            printf("Department: %s\n", employees[i].department);
```

```
            printf("Position: %s\n", employees[i].position);
            printf("Basic Salary: %.2f\n", employees[i].basicSalary);
            printf("Allowance: %.2f\n", employees[i].allowance);
            printf("Deduction: %.2f\n", employees[i].deduction);

            printf("============================\n");

            found = 1;
            break;
        }
    }

    if (found == 0) {
        printf("\nEmployee not found.\n");
    }
}
```